

Please check the examination details below before entering your candidate information

|                                                                                                                                                        |                                                                                                          |                                                                                     |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------|
| Candidate surname                                                                                                                                      |                                                                                                          | Other names                                                                         |             |
| <b>Pearson Edexcel</b><br><b>International</b><br><b>Advanced Level</b>                                                                                | Centre Number                                                                                            | Candidate Number                                                                    |             |
|                                                                                                                                                        | <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> | <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> |             |
| <b>Tuesday 12 January 2021</b>                                                                                                                         |                                                                                                          |                                                                                     |             |
| Morning (Time: 1 hour 30 minutes)                                                                                                                      |                                                                                                          | Paper Reference <b>WCH11/01</b>                                                     |             |
| <b>Chemistry</b><br><b>International Advanced Subsidiary/Advanced Level</b><br><b>Unit 1: Structure, Bonding and Introduction to Organic Chemistry</b> |                                                                                                          |                                                                                     |             |
| <b>You must have:</b><br>Scientific calculator, Ruler                                                                                                  |                                                                                                          |                                                                                     | Total Marks |

## Instructions

- Use **black** ink or **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided  
– *there may be more space than you need.*
- Show all your working in calculations and include units where appropriate.

## Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*
- You will be assessed on your ability to organise and present information, ideas, descriptions and arguments clearly and logically, including your use of grammar, punctuation and spelling.
- A Periodic Table is printed on the back cover of this paper.

## Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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## SECTION A

Answer ALL questions. Write your answers in the spaces provided.

For each question, select one answer from A to D and put a cross in the box ☐. If you change your mind, put a line through the box ☐ and then mark your new answer with a cross ☐.

1 Which of these compounds has the same empirical and molecular formulae?

- ☐ A  $\text{C}_2\text{H}_4$
- ☐ B  $\text{C}_3\text{H}_8$
- ☐ C  $\text{C}_4\text{H}_{10}$
- ☐ D  $\text{C}_5\text{H}_{10}$

(Total for Question 1 = 1 mark)

2 There are  $6.02 \times 10^{23}$  atoms in 0.25 mol of

- ☐ A He
- ☐ B  $\text{H}_2\text{O}$
- ☐ C  $\text{BH}_3$
- ☐ D  $\text{CH}_4$

(Total for Question 2 = 1 mark)

3 Which aqueous solution has the **highest** concentration, in  $\text{mol dm}^{-3}$ , of chloride ions?

- ☐ A  $0.1 \text{ g dm}^{-3} \text{ HCl}$
- ☐ B  $0.1 \text{ g dm}^{-3} \text{ NaCl}$
- ☐ C  $0.1 \text{ g dm}^{-3} \text{ KCl}$
- ☐ D  $0.1 \text{ g dm}^{-3} \text{ BaCl}_2$

(Total for Question 3 = 1 mark)

4 Which conversion has the **lowest** percentage atom economy (by mass) for the formation of  $\text{CaCl}_2$ ?

- ☐ A  $\text{Ca} + \text{Cl}_2 \rightarrow \text{CaCl}_2$
- ☐ B  $\text{Ca} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$
- ☐ C  $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
- ☐ D  $\text{CaCO}_3 + 2\text{NaCl} \rightarrow \text{CaCl}_2 + \text{Na}_2\text{CO}_3$

(Total for Question 4 = 1 mark)



5 Which of these atoms has the most neutrons?

- ☐ A  $^{115}_{49}\text{In}$
- ☐ B  $^{124}_{50}\text{Sn}$
- ☐ C  $^{123}_{51}\text{Sb}$
- ☐ D  $^{124}_{52}\text{Te}$

(Total for Question 5 = 1 mark)

6 A p-block element in **Period 3** of the Periodic Table reacts to form an ionic compound.

What could be the electronic configuration of the **ion** formed by this element?

- ☐ A  $1s^2 2s^2 2p^6 3s^2$
- ☐ B  $1s^2 2s^2 2p^6 3s^2 3p^6$
- ☐ C  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$
- ☐ D  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

(Total for Question 6 = 1 mark)

7 The **Period 2** element with the **highest** melting temperature is

- ☐ A aluminium
- ☐ B boron
- ☐ C carbon
- ☐ D silicon

(Total for Question 7 = 1 mark)

8 Which of these has the **greatest** electrical conductivity?

- ☐ A  $\text{SF}_6(\text{g})$
- ☐ B  $\text{H}_2\text{O}(\text{l})$
- ☐ C  $\text{Hg}(\text{l})$
- ☐ D  $\text{Na}_2\text{O}(\text{s})$

(Total for Question 8 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.



9 Which of these ions has the **greatest** ionic radius?

- ☐ A  $\text{N}^{3-}$
- ☐ B  $\text{F}^-$
- ☐ C  $\text{Na}^+$
- ☐ D  $\text{Al}^{3+}$

(Total for Question 9 = 1 mark)

10 Which of these ions has the **greatest** polarising power?

- ☐ A  $\text{S}^{2-}$
- ☐ B  $\text{Cl}^-$
- ☐ C  $\text{K}^+$
- ☐ D  $\text{Ca}^{2+}$

(Total for Question 10 = 1 mark)

11 Which of these does **not** have a structure formed by a giant lattice of carbon atoms?

- ☐ A  $\text{C}_{60}$  fullerene
- ☐ B diamond
- ☐ C graphene
- ☐ D graphite

(Total for Question 11 = 1 mark)

12 Which of these molecules is the **most** polar?

- ☐ A HF
- ☐ B  $\text{OF}_2$
- ☐ C  $\text{BF}_3$
- ☐ D  $\text{CF}_4$

(Total for Question 12 = 1 mark)

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13 A substance is labelled with the hazard symbol shown.

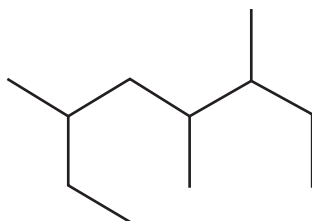


What is the meaning of this symbol?

- ☐ A gloves must be worn
- ☐ B corrosive
- ☐ C do not store with flammable substances
- ☐ D oxidising

(Total for Question 13 = 1 mark)

14 What is the IUPAC name of this alkane?



- ☐ A 2-ethyl-4,5-dimethylheptane
- ☐ B 6-ethyl-3,4-dimethylheptane
- ☐ C 3,4,6-trimethyloctane
- ☐ D 3,5,6-trimethyloctane

(Total for Question 14 = 1 mark)

15 Petrol, bioethanol and hydrogen are fuels.

All three of these fuels

- ☐ A burn to produce greenhouse gases
- ☐ B are overall carbon neutral
- ☐ C are overall sustainable
- ☐ D biodegrade rapidly

(Total for Question 15 = 1 mark)



16 Cyclopentane undergoes free radical substitution with bromine.

(a) Which of these is an overall equation for this reaction?

(1)

- ☐ A  $C_5H_8 + Br_2 \rightarrow C_5H_8Br_2$
- ☐ B  $C_5H_{10} + Br_2 \rightarrow C_5H_{10}Br_2$
- ☐ C  $C_5H_{10} + Br_2 \rightarrow C_5H_8Br_2 + H_2$
- ☐ D  $C_5H_{10} + Br_2 \rightarrow C_5H_9Br + HBr$

(b) Which statement is **not** correct about this reaction system?

(1)

- ☐ A only the initiation step involves homolytic bond fission
- ☐ B only some bromine is converted to free radicals in the initiation step
- ☐ C propagation forms more product than termination
- ☐ D further substitution reactions are likely to occur

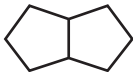
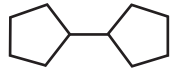
(c) Which free radical is **least** likely to form in a propagation step in this reaction system?

(1)

- ☐ A  $C_5H_9^\bullet$
- ☐ B  $Br^\bullet$
- ☐ C  $C_5H_8Br^\bullet$
- ☐ D  $H^\bullet$

(d) Which alkane could be formed in a termination step in this reaction system?

(1)

- ☐ A 
- ☐ B 
- ☐ C 
- ☐ D 

(Total for Question 16 = 4 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.



17 Which of these is **not** a way of limiting global problems caused by polymer disposal?

- ☐ A developing biodegradable polymers
- ☐ B exporting polymer waste
- ☐ C removing toxic waste gases produced by the incineration of polymers
- ☐ D reusing products made from polymers

(Total for Question 17 = 1 mark)

**TOTAL FOR SECTION A = 20 MARKS**



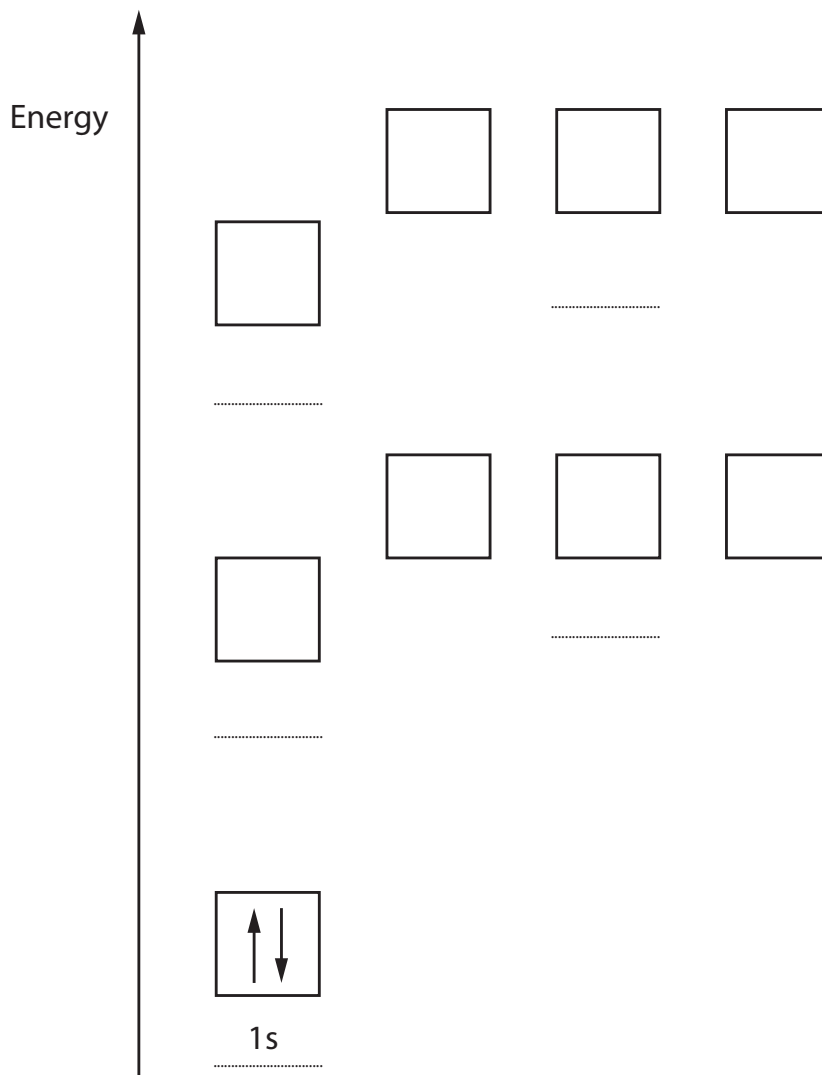
## SECTION B

Answer ALL the questions. Write your answers in the spaces provided.

18 This question is about the element sulfur.

- (a) Complete the diagram to show the electronic configuration for a sulfur atom in the ground state.  
Include labels for each subshell.

(2)



- (b) Write an equation for the **first** ionisation energy of sulfur.  
Include state symbols.

(2)

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- (c) Explain why the first ionisation energy of sulfur is less than the first ionisation energies of **both** phosphorus and chlorine.

(3)

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- (d) A sample of sulfur contains four isotopes.

| Isotope              | $^{32}\text{S}$ | $^{33}\text{S}$ | $^{34}\text{S}$ | $^{36}\text{S}$ |
|----------------------|-----------------|-----------------|-----------------|-----------------|
| Percentage abundance | 94.88           | 0.83            | 4.27            | 0.02            |

- (i) State what is meant by the term **isotopes**, in terms of subatomic particles.

(2)

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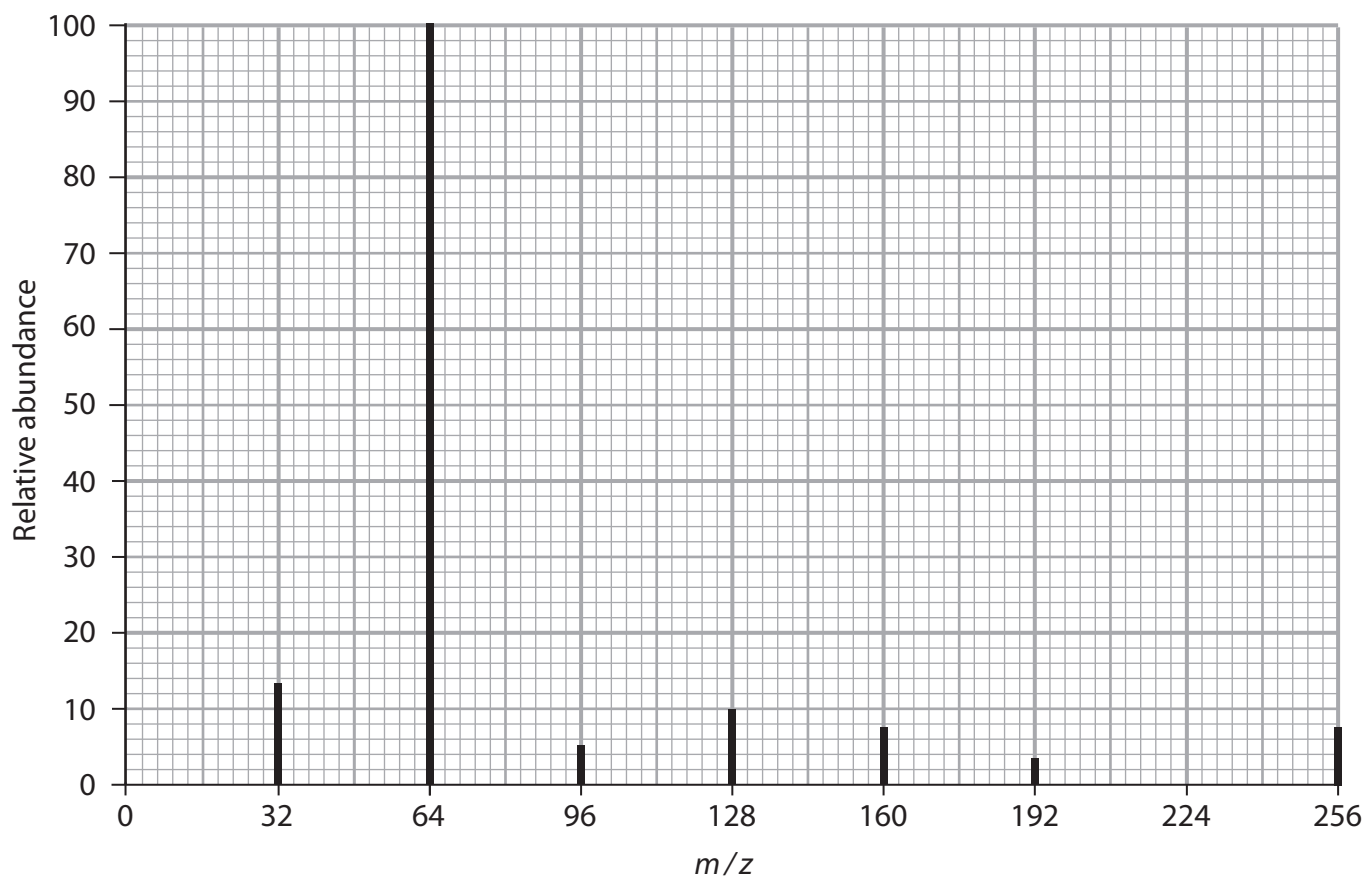
- (ii) Calculate the relative atomic mass of sulfur in this sample.

Give your answer to **two** decimal places.

(2)



(e) The mass spectrum of a sample of sulfur with  $^{32}\text{S}$  as the only isotope is shown.



(i) Calculate the number of sulfur atoms in the molecular ion.

You **must** show your working.

(1)

(ii) Suggest the **formula** of the **most stable ion** shown by this spectrum.

(2)

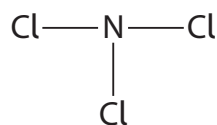
(Total for Question 18 = 14 marks)



**19** This question is about the structure and bonding of Group 5 chlorides.

- (a) Nitrogen trichloride,  $\text{NCl}_3$ , has a molecular structure.

The displayed formula of a molecule of  $\text{NCl}_3$  is shown.



Complete the table for this molecule.

(3)

|                                    |  |
|------------------------------------|--|
| Number of bond pairs around N atom |  |
| Number of lone pairs around N atom |  |
| Cl-N-Cl bond angle                 |  |
| Name of shape of molecule          |  |

- (b) Under standard conditions, phosphorus(V) chloride ( $\text{PCl}_5$ ) is a solid made up of  $\text{PCl}_4^+$  cations and  $\text{PCl}_6^-$  anions.

Antimony(V) chloride ( $\text{SbCl}_5$ ) is a liquid made up of  $\text{SbCl}_5$  molecules.

- (i) Explain why  $\text{PCl}_5$  has a higher melting temperature than  $\text{SbCl}_5$ .

(2)

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(ii) Draw a dot-and-cross diagram to show the bonding in a molecule of  $\text{SbCl}_5$ .

Use dots (•) to represent the Sb electrons, and crosses (x) to represent the Cl electrons. Show outer electrons only.

(2)

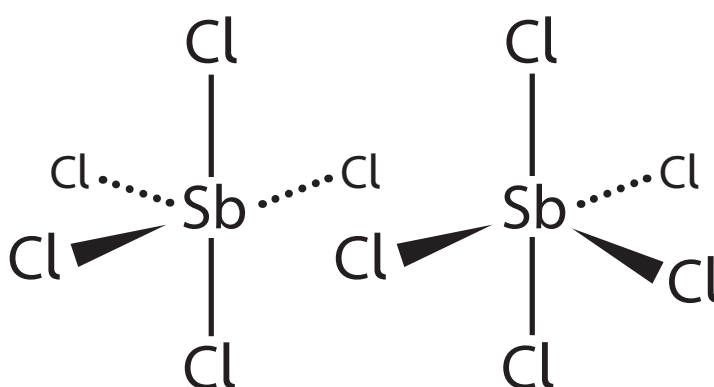
(c) At low temperatures,  $\text{SbCl}_5$  converts to  $\text{Sb}_2\text{Cl}_{10}$  which contains dative covalent bonds.

(i) State what is meant by the term dative covalent bond.

(1)

(ii) Complete the diagram to show the dative covalent bonds in  $\text{Sb}_2\text{Cl}_{10}$ .

(1)



(d) Arsenic also forms a pentachloride with the formula  $\text{AsCl}_5$ .

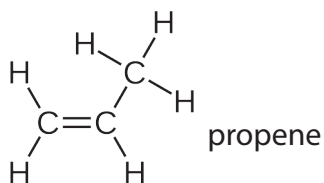
Give **one** possible reason why nitrogen is the only Group 5 element that does **not** form a pentachloride.

(1)

(Total for Question 19 = 10 marks)



20 This question is about the reactions of propene.



- (a) Write an equation for the incomplete combustion of **one mole** of propene to form carbon dioxide, carbon monoxide, carbon and water as the only products. Include state symbols.

(2)

- (b) State **one** similarity and **one** difference that would be **seen** when propene is mixed with separate samples of acidified potassium manganate(VII) solution and of bromine water.

(2)

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- (c) Propene reacts by addition polymerisation to form poly(propene).

Draw the structure of poly(propene), showing **two** repeat units.

(1)



(d) Propene reacts with bromine monochloride, BrCl, to form 1-bromo-2-chloropropane as the major product.

(i) Complete the diagram of bromine monochloride to show the dipole.

(1)



(ii) Draw the mechanism for the formation of 1-bromo-2-chloropropane in this reaction.  
Include curly arrows, and relevant lone pairs.

(3)

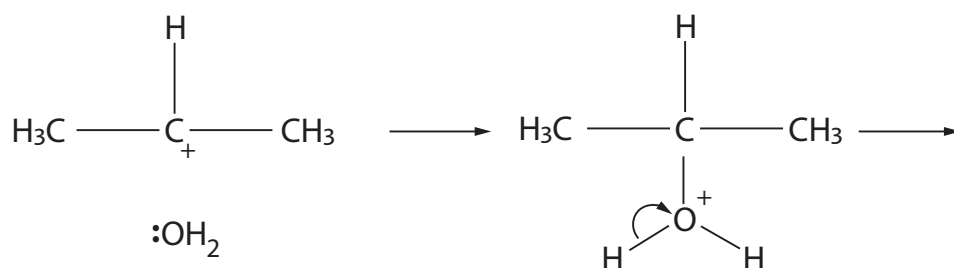
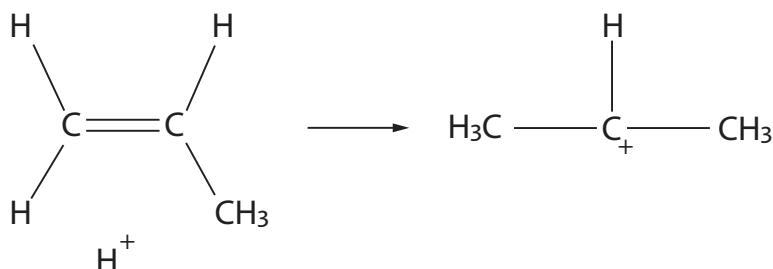


- (e) Propene reacts with steam in the presence of an acid catalyst to form a mixture of the alcohols propan-1-ol and propan-2-ol.

Complete the mechanism for the formation of propan-2-ol, by adding curly arrows.

Include the species formed in the final step.

(3)



(Total for Question 20 = 12 marks)





- (iii) A reactor at 200°C contains 8500 mol of liquid squalene, and hydrogen gas at a pressure of  $4.0 \times 10^5$  Pa.

Under these conditions, the complete hydrogenation of squalene requires 500 m<sup>3</sup> of hydrogen gas.

Calculate the number of C=C bonds in one molecule of squalene.

You **must** show your working.

$$[pV = nRT \quad R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}]$$

(4)

- (iv) Write the equation, using molecular formulae, for the complete hydrogenation of squalene to squalane.  
State symbols are **not** required.

(1)



- (d) Globally, 2.8 million  $\text{dm}^3$  of squalene is used each year.  
Traditionally squalene was obtained exclusively from shark liver oil, which is a mixture of liquids.

The liver of a shark yields 300 g of squalene.

- (i) Suggest the name of a suitable technique to obtain squalene from shark liver oil.

(1)

- (ii) Calculate the minimum number of sharks that would be needed to produce 2.8 million  $\text{dm}^3$  of squalene.

[Density of squalene =  $0.86 \text{ g cm}^{-3}$ ]

(2)

- (iii) Many large corporations now use squalane obtained entirely from plants.

Squalane can be obtained sustainably from corn starch with a yield of 23 % by mass.

The production of 1 tonne of corn starch requires 0.093 hectares of land.

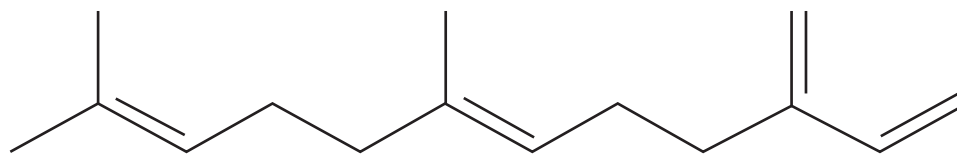
Calculate the area of land, **in  $\text{km}^2$** , required to produce 2500 tonnes of squalane from corn starch.

[1 tonne = 1000 kg                      1 hectare =  $0.01 \text{ km}^2$ ]

(3)



(e) The *E*-isomer of beta-farnesene can also be obtained from corn starch.



*E*-beta-farnesene

- (i) Explain why beta-farnesene exhibits geometric isomerism and has only two geometric isomers.

You may label the structure and use this in your answer.

(2)

- (ii) Draw the **skeletal** formula of the geometric isomer of *E*-beta-farnesene, giving a reason why this is named the *Z*-isomer.

(2)



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## 24

P 6 7 7 5 2 A 0 2 4 2 4

\* Lanthanide series

\* Actinide series

|     |    |         |    |     |    |              |    |     |    |           |    |     |    |            |    |     |    |           |    |     |    |           |    |     |    |            |    |     |    |           |    |     |    |             |    |     |    |             |    |     |    |         |     |     |    |              |     |     |    |           |     |     |    |            |     |
|-----|----|---------|----|-----|----|--------------|----|-----|----|-----------|----|-----|----|------------|----|-----|----|-----------|----|-----|----|-----------|----|-----|----|------------|----|-----|----|-----------|----|-----|----|-------------|----|-----|----|-------------|----|-----|----|---------|-----|-----|----|--------------|-----|-----|----|-----------|-----|-----|----|------------|-----|
| 140 | Ce | cerium  | 58 | 141 | Pr | praseodymium | 59 | 144 | Nd | neodymium | 60 | 147 | Pm | promethium | 61 | 150 | Sm | samarium  | 62 | 152 | Eu | europium  | 63 | 157 | Gd | gadolinium | 64 | 159 | Tb | terbium   | 65 | 163 | Dy | dysprosium  | 66 | 165 | Ho | holmium     | 67 | 167 | Er | erbium  | 68  | 169 | Tm | thulium      | 69  | 173 | Yb | ytterbium | 70  | 175 | Lu | lutetium   | 71  |
| 232 | Th | thorium | 90 | 231 | Pa | protactinium | 91 | 238 | U  | uranium   | 92 | 237 | Np | neptunium  | 93 | 242 | Pu | plutonium | 94 | 243 | Am | americium | 95 | 247 | Cm | curium     | 96 | 245 | Bk | berkelium | 97 | 251 | Cf | californium | 98 | 254 | Es | einsteinium | 99 | 253 | Fm | fermium | 100 | 256 | Md | mendeleevium | 101 | 254 | No | nobelium  | 102 | 257 | Lr | lawrencium | 103 |

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